Summary Notes:
DevOps On Azure

Overview

Here is an overview of the topics that we will cover:

1. Session 1
    a. What is DevOps?
    b. Key Benefits of DevOps
    c. DevOps Practices
    d. Git Basics
    e. Benefits of Git
    f. What is Azure DevOps
    g. Key Components of Azure DevOps

2. Session 2
    a. What are Delivery Plans
    b. Key Features of Delivery Plans
    c. Benefits of Delivery Plans

Session 1: Azure DevOps and Its Features

What is DevOps?

DevOps integrates development (Dev) and operations (Ops) to enhance collaboration between traditionally siloed teams, including IT operations, quality engineering, security, and development.

Key Benefits of DevOps:

- Faster time to market.
- Greater adaptability to market needs and competition.
- Improved system stability and reliability.
- Reduced mean time to recovery.

DevOps Practices:

- **Continuous Integration/Continuous Delivery (CI/CD)**: Automates build, test, and deployment workflows.
- **Version Control**: Tracks and manages code changes for better collaboration.
- **Infrastructure as Code (IaC)**: Manages infrastructure programmatically for consistency and repeatability.
- **Agile Development**: Promotes incremental delivery and continuous feedback.

Git Basics

Git is a distributed version control system enabling seamless collaboration and offline work.

Core Concepts:

1. **Commits**: Snapshots of all file changes linked to form a history, with unique IDs for tracking.
2. **Branches**: Parallel lines of development that can be merged into the main branch.
3. **Staging and Committing**: Changes are staged before being committed to the project history.

Benefits of Git:

- Facilitates simultaneous development and faster releases.

- Offers built-in integration and robust community support.
- Features pull requests and branch policies for collaborative workflows.

What is Azure DevOps?

Azure DevOps is a cloud-based suite of tools designed to streamline collaboration, automation, and software delivery. It supports both cloud and on-premises implementations.

Key Components:

1. **Azure Boards**:
    - Facilitates planning and tracking with Agile tools like Kanban boards and backlogs.
    - Visualizes workflows for better progress tracking.
2. **Azure Repos**:
    - A robust version control system supporting Git and TFVC.
    - Features include branch policies, pull requests, and offline work with repositories.
3. **Azure Pipelines**:
    - Automates build, test, and deployment workflows.
    - Supports CI/CD, build automation, and release tracking with approval workflows.
4. **Azure Test Plans**:
    - Enables manual, exploratory, and automated testing for quality assurance.
    - Provides traceability and collaboration tools.
5. **Azure Artifacts**:
    - Securely manages and shares package dependencies like Maven, npm, and NuGet.
    - Tracks external package usage and ensures traceability.

Additional Features:

- **Dashboards**: Customizable views for tracking workflows and metrics.
- **Service Hooks**: Automates integration with external services for quick responses.
- **Azure DevOps Server**: On-premises solutions with SQL Server and GitHub integration.

Session 2: Delivery Plans

What are Delivery Plans?

Delivery Plans provide a clear, real-time visualization of team schedules, improving alignment and decision-making.

Key Features:

1. **Visualization of Team Workloads**: Dynamically updated schedules based on team backlogs.
2. **Dependency Tracking**: Highlights dependencies across teams and projects.
3. **Proactive Conflict Resolution**: Regular reviews address scheduling conflicts and gaps.

Benefits of Delivery Plans:

- Aligns team efforts with organizational goals.
- Enhances visibility into project schedules and dependencies.
- Boosts delivery confidence through proactive monitoring.